Submission Summary

Conference Name
IEEE Chandigarh Subsection International Conference

Track Name
Artificial Intelligence and Data Science

Paper ID
873

Paper Title
Machine Learning Based Drug-Drug Interaction Prediction: A TWOSIDES Dataset and Ensemble Classifier Approach

Abstract
When you take two medicines at the time they do not always work the way they should. The body breaks down medicines using things called enzymes and pathways. When two medicines use the same one it can be bad. These things are called drug-drug interactions and hospitals and pharmacists have to deal with them every day. They have to find these interactions before they cause problems.This project is trying to build a computer program that can predict whether two medicines will interact or not. It uses data from the TWOSIDES dataset to do this. The computer program looks at pairs of medicines. Tries to figure out if they will interact. Each pair of medicines is turned into a set of numbers that the computer can understand.The project uses four computer models to try to predict interactions. These models are Logistic Regression, Random Forest, XGBoost and a special kind of network. The XGBoost model was pretty good at predicting interactions. The neural network was even better. This is an improvement over other work that used fake data. The project found out that medicines that are similar in structure are more likely to interact. This makes sense because medicines that are similar are often broken down by the enzymes in the body. The project also includes a web interface where you can test the computer model with any two medicines. You just need to type in the SMILES strings, for the two medicines.

Created
2026-04-26 13:40:11 GMT+5:30

Last Modified
2026-04-26 13:40:11 GMT+5:30

Authors
Paarth Sharma (Chitkara University) <paarth630.be22@chitkara.edu.in>

Submission Files
DDI_Research_Paper.docx (27.4 Kb, 2026-04-26 13:38:53 GMT+5:30)

Submission Questions Response
1. Corresponding author Email ID
Provide Email ID of corresponding author

paarth630.be22@chitkara.edu.in

2. Contact number with country code

Provide Whatsapp number of corresponding author with country code

+91 9872003172

3. Author's Information

I confirm that I have listed the title of the paper and the names, email addresses and affiliations of all contributed authors in the submission system that match the order as given in the submitted paper

Agreement accepted

4. Co-author's Consent

I confirm that I have discussed about this submission with all co-authors and authors have approved the paper and agree with its submission to this conference.

Agreement accepted

5. Certificate of Originality

I confirm that neither the paper nor any parts of its content are currently under consideration or published in another conference/journal and the paper is free from plagiarism

Agreement accepted

6. Author's Consent

It is confirmed that at least one author will register in full to attend and present the paper at the Conference.

Agreement accepted

7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

I confirm that no text in the paper is AI-generated.

Agreement accepted